Linear Algebra 1A Cheat Sheet

Convention: a matrix of size $r \times c$ has r rows and c columns.

1. Fields

A field $\mathbb{F}$ is a set with operations $(+, \cdot)$ satisfying: closure, associativity, commutativity, identities $(0, 1)$, inverses, and distributivity.

Characteristic.

- $\mathrm{char}(\mathbb{F}) :=$ the least $n \geq 1$ with $\overbrace{1 + \cdots + 1}^{n \text{ times}} = 0$, and $\mathrm{char}(\mathbb{F}) := 0$ if no such n exists
- $\mathrm{char}(\mathbb{F})$ is always 0 or prime
- $\forall \mathbb{F} : \mathrm{char}(\mathbb{F}) = p > 0 \Rightarrow \mathbb{Z}_p$ is a subfield of $\mathbb{F}$

Finite fields.

- $|\mathbb{F}| < \infty \Rightarrow |\mathbb{F}| = p^r$, where $p = \mathrm{char}(\mathbb{F})$ and $r \in \mathbb{N}$
- $\forall (p \in \mathbb{P}) \,.\, \forall (r \in \mathbb{N}) \,.\, \exists (\mathbb{F} \text{ field}) : |\mathbb{F}| = p^r$

2. Matrices

$$A_{m \times p} B_{p \times n} = (AB)_{m \times n}, \qquad (AB)_{ij} := \sum_{k=1}^{p} a_{ik} b_{kj}$$

- B with columns $b_1, \ldots, b_n$: $AB = (Ab_1, \ldots, Ab_n)$
- A with rows $a_1, \ldots, a_m$: row i of AB is $a_i B$
- $AB = 0$ does not imply $A = 0$ or $B = 0$
- If $A^T = A$ and $B^T = B$, then AB is symmetric $\iff AB = BA$
- If φ is an elementary row operation and $A \in \mathbb{F}^{m \times n}$, then

  $$\varphi(A) = \varphi(I_m)\, A$$

3. Vector Spaces & Subspaces

A subset $W \subseteq V$ is a subspace if and only if:

- W is nonempty (equivalently $\vec{0} \in W$)
- Closed under addition: if $u, v \in W$ then $u + v \in W$
- Closed under scalar multiplication: if $\alpha \in \mathbb{F}$ and $u \in W$ then $\alpha u \in W$

Equivalently: $\alpha u + v \in W$ for all $u, v \in W$ and all $\alpha \in \mathbb{F}$.

- $U \cap W$ is a subspace of V
- $U \cup W$ is a subspace of $V \iff U \subseteq W$ or $W \subseteq U$
- $U + W := \{u + w : u \in U,\ w \in W\} = \mathrm{span}(U \cup W)$ is a subspace of V
- Direct sum: $U \oplus W \iff U \cap W = \{\vec{0}\}$ (general case: see Direct sums, §6)
- $\mathrm{span}(S)$ is the smallest subspace containing S
- $S_1 \subseteq S_2 \subseteq V \Rightarrow \mathrm{span}(S_1) \leq \mathrm{span}(S_2) \leq V$
- $W \leq V \Rightarrow \mathrm{span}(W) = W$
- $S \subseteq W \leq V \Rightarrow \mathrm{span}(S) \leq W$

4. Rank

- $\mathrm{rank}(A) :=$ the number of pivots in $\mathrm{RREF}(A)$
- $\mathrm{rank}(A_{m \times n}) \leq \min\{m, n\}$
- $\mathrm{rank}(A^T) = \mathrm{rank}(A)$
- $A \sim B \Rightarrow \mathrm{rank}(A) = \mathrm{rank}(B)$
- $\mathrm{rank}(AB) \leq \min\{\mathrm{rank}(A), \mathrm{rank}(B)\}$
- $A, B \in \mathbb{F}^{n \times n},\ AB = 0 \Rightarrow \mathrm{rank}(A) + \mathrm{rank}(B) \leq n$

5. Linear Systems

$Ax = b, \qquad n$ variables, $\qquad r = \mathrm{rank}(A)$

- Number of pivots $= r$, free variables $= n - r$
- System is solvable $\iff b \in \mathrm{Col}(A)$
- x_p: any one solution of $Ax = b$ (a *particular* solution)
- to find x_p: reduce $[A \mid b]$, set every free variable to 0, then
  - from RREF: read off the pivot variables
  - from row echelon form: back-substitute
- Homogeneous system: $Ax = \vec{0}$ has solution space $\mathrm{Null}(A) = \ker(A)$, of dimension $n - r$
- $\{x : Ax = b\} = x_p + \ker(A)$

6. Bases & Dimension

If $\dim V = n$:

- Any linearly independent set has size at most n
- Any spanning set has size at least n
- $\dim V < \infty$: any linearly independent set extends to a basis of V
- $\dim V < \infty$: any spanning set contains a basis of V
- A set B with $|B| = n$ is a basis $\iff$ it is linearly independent $\iff$ it spans V
- If $W \leq V$, then $W = V \iff \dim W = \dim V$

$\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$

Direct sums. Let $M_1, \ldots, M_s \leq V$ with $M = M_1 + \cdots + M_s$. The following are equivalent:

1. $M = M_1 \oplus \cdots \oplus M_s$
2. $\forall a \in M$: the representation $a = a_1 + \cdots + a_s$, $a_i \in M_i$, is unique
3. $\exists a \in M$: the representation $a = a_1 + \cdots + a_s$, $a_i \in M_i$, is unique
4. $\forall a_i \in M_i\ :\ a_1 + \cdots + a_s = \vec{0} \Rightarrow a_1 = \cdots = a_s = \vec{0}$
5. $E^{(i)}$ a basis of $M_i \Rightarrow E^{(1)}, \ldots, E^{(s)}$ together form a basis of M
6. $\dim M = \dim M_1 + \cdots + \dim M_s$

Note 3 $\Rightarrow$ all: one witness with a unique decomposition suffices.

7. Coordinates

Let $B = \{v_1, \ldots, v_n\}$ be a basis of V.

- $[B] := (v_1, \ldots, v_n)$, the basis written as a row of its vectors
- Every $v \in V$ has a unique representation $v = \alpha_1 v_1 + \cdots + \alpha_n v_n$
- Coordinate vector: $[v]_B := (\alpha_1, \ldots, \alpha_n)^T$
- $v = [B]\, [v]_B$
- $u_1, \ldots, u_n$ linearly independent, $A \in \mathbb{F}^{n \times p}$: $(u_1, \ldots, u_n)\, A = (\vec{0}, \ldots, \vec{0}) \Rightarrow A = 0$

Change of basis. The transition matrix from basis B to basis C, written $[I]_{B \to C}$ (also $[I]_B^C$), has the $[v_i]_C$ as its columns:

- $[I]_{B \to C} := \big([v_1]_C, \ldots, [v_n]_C\big)$
- $[v]_C = [I]_{B \to C}\, [v]_B$

- B, C, D bases: $[I]_{B\to D} = [I]_{C\to D}\,[I]_{B\to C}$
- $[I]_{B\to C}$ is invertible, with $\left([I]_{B\to C}\right)^{-1} = [I]_{C\to B}$
- $[C] = (w_1, \dots, w_n)$: $[C] = [B]\,P \Rightarrow P = [I]_{C\to B}$

8. Invertible Matrices

- P invertible $\Rightarrow \operatorname{rank}(PA) = \operatorname{rank}(AP) = \operatorname{rank}(A)$
- $\alpha \neq 0 \Rightarrow (\alpha A)^{-1} = \alpha^{-1} A^{-1}$
- C invertible, $B = CA$: B invertible $\iff$ A invertible

Conditions Related to Invertibility

Let $A \in \mathbb{F}^{n\times n}$. Each item states its relation to "A is invertible":

1. $\implies$ A^{-1} is unique
2. $\implies$ A has no zero row or column
3. $\iff$ $\operatorname{rank}(A) = n$
4. $\iff$ $\det(A) \neq 0$
5. $\iff$ A^T is invertible, with $(A^T)^{-1} = (A^{-1})^T$
6. $\iff$ $A \sim I_n$
7. $\iff$ $\operatorname{RREF}(A) = I_n$
8. $\iff$ $Ax = \vec{0} \Rightarrow x = \vec{0}$
9. $\iff$ For all $b \in \mathbb{F}^n$, the system $Ax = b$ has the unique solution $x = A^{-1}b$
10. $\iff$ $\operatorname{Row}(A) = \mathbb{F}^n$
11. $\iff$ $\operatorname{Col}(A) = \mathbb{F}^n$
12. $\iff$ The rows of A span $\mathbb{F}^n$
13. $\iff$ The columns of A span $\mathbb{F}^n$
14. $\iff$ The rows of A are linearly independent
15. $\iff$ The columns of A are linearly independent
16. $\iff$ The rows of A form a basis of $\mathbb{F}^n$
17. $\iff$ The columns of A form a basis of $\mathbb{F}^n$
18. $\iff$ $\dim \operatorname{Row}(A) = n$
19. $\iff$ $\dim \operatorname{Col}(A) = n$
20. $\iff$ A is a product of elementary matrices
21. $\iff$ $\operatorname{adj}(A)$ is invertible
22. $\iff$ A is invertible on one side (i.e. $AB = I$ or $BA = I$ for some B) — **square A only**
23. $\iff$ $A = [I]_{B\to C}$ for some bases B, C
24. $\impliedby$ A is elementary (then A^{-1} is also elementary)

9. Matrix Spaces

$A \in \mathbb{F}^{m\times n}$

- $\operatorname{Row}(A) := \operatorname{span}(\text{rows of } A) \leq \mathbb{F}^n$
- $\operatorname{Col}(A) := \operatorname{span}(\text{columns of } A) \leq \mathbb{F}^m$
- $\dim \operatorname{Row}(A) = \dim \operatorname{Col}(A) = \operatorname{rank}(A)$
- **ROW OPERATIONS DO NOT PRESERVE THE COLUMN SPACE**
  - $\operatorname{Row}(A)$ is preserved
  - $\dim \operatorname{Col}(A) = \operatorname{rank}(A)$ is preserved, $\operatorname{Col}(A)$ is not
- $\operatorname{Null}(A) = \ker(A) = \{x : Ax = \vec{0}\}$
- $\dim \ker(A) = n - \operatorname{rank}(A)$
- $\operatorname{Row}(AB) \subseteq \operatorname{Row}(B)$
- $\operatorname{Col}(AB) \subseteq \operatorname{Col}(A)$

10. Determinants & Adjugate

- $\det(A^T) = \det(A)$
- $\det(AB) = \det(A)\det(B)$
- $\det(A^{-1}) = (\det A)^{-1}$
- $\det(\alpha A) = \alpha^n \det(A)$ for $A \in \mathbb{F}^{n\times n}$

Adjugate.
$A \in \mathbb{F}^{n\times n}$

- $M_{ij}(A) :=$ A with row i and column j deleted
- cofactor $c_{ij} := (-1)^{i+j}\det(M_{ij}(A))$
- $\operatorname{adj}(A)_{ij} := c_{ji} = (-1)^{i+j}\det(M_{ji}(A))$
- $\operatorname{adj}(A)\,A = A\,\operatorname{adj}(A) = \det(A) I_n$
- $A^{-1} = \frac{1}{\det(A)}\operatorname{adj}(A)$
- $\operatorname{rank}(\operatorname{adj}(A)) = \begin{cases} n, & \operatorname{rank}(A) = n, \\ 1, & \operatorname{rank}(A) = n-1, \\ 0, & \operatorname{rank}(A) < n-1 \end{cases}$

Cramer's rule. $\det(A) \neq 0 \Rightarrow Ax = b$ has the unique solution

$$x_k = \frac{\det(A_k)}{\det(A)}, \qquad k = 1, \dots, n$$

where A_k is A with column k replaced by b.

11. Linear Maps

$T : V \to U$ is linear if for all $u, v \in V$ and $\alpha \in \mathbb{F}$:

- $T(u + v) = T(u) + T(v)$
- $T(\alpha v) = \alpha T(v)$

Equivalently: $T(\alpha u + v) = \alpha T(u) + T(v)$.

- $\{v_1, \dots, v_n\}$ a basis of V: any choice of $T(v_1), \dots, T(v_n) \in U$ determines a unique linear $T : V \to U$
- T_1, T_2 linear $\Rightarrow T_1 \circ T_2$ is linear
- $\{v_1, \dots, v_s\}$ linearly dependent $\Rightarrow \{T(v_1), \dots, T(v_s)\}$ linearly dependent
- Kernel: $\ker T := \{v \in V : T(v) = \vec{0}\}$
  - $\ker T \leq V$
- Image: $\operatorname{Im} T := \{T(v) : v \in V\}$
  - $\operatorname{Im} T \leq U$
- $\dim V < \infty$, $\{e_1, \dots, e_k\}$ a basis of $\ker T$ extended to a basis $\{e_1, \dots, e_n\}$ of V $\Rightarrow \{T(e_{k+1}), \dots, T(e_n)\}$ is a basis of $\operatorname{Im} T$

$\dim V = \dim \ker T + \dim \operatorname{Im} T$

12. Operators & Isomorphisms

Let $T : V \to U$ be linear.

- Monomorphism := injective T
  - T_1, T_2 mono $\Rightarrow T_1 \circ T_2$ mono
- Epimorphism := surjective T
  - T_1, T_2 epi $\Rightarrow T_1 \circ T_2$ epi
- Isomorphism := bijective T; write $V \cong U$ if one exists
  - T_1, T_2 iso $\Rightarrow T_1 \circ T_2$ iso
  - T iso $\Rightarrow T^{-1}$ iso
  - $\cong$ is an equivalence relation
  - $\dim V, \dim U < \infty$: T iso $\iff$ T maps every basis of V to a basis of U $\iff$ T maps some basis of V to a basis of U
- T injective $\iff$ $\ker T = \{\vec{0}\}$
- T surjective $\iff$ $\dim \operatorname{Im} T = \dim U$
- $V \cong U \iff \dim V = \dim U$
- $\dim V = \dim U < \infty$: T mono $\iff$ T epi $\iff$ T iso

- For an operator $T : V \to V$:

$$\ker T \subseteq \ker T^2 \subseteq \dots, \qquad \operatorname{Im} T \supseteq \operatorname{Im} T^2 \supseteq \dots$$

  - $\ker T^k = \ker T^{k+1} \Rightarrow \ker T^m = \ker T^k$ for all $m \geq k$ (same for Im)
  - $\dim V < \infty \Rightarrow$ this equality occurs, by $k = \dim V$ at the latest

13. Matrix Representation

$T : V \to W$ linear
$B = \{v_1, \dots, v_n\}$ basis of V
$C = \{w_1, \dots, w_m\}$ basis of W
$[\varphi]_B := [\varphi]_{B\to B}$ for $\varphi : V \to V$

- $[T]_{B\to C} := \big([T(v_1)]_C, \dots, [T(v_n)]_C\big) \in \mathbb{F}^{m\times n}$
- $[T]_{B\to C}\,[v]_B = [T(v)]_C$ for all $v \in V$
- $\operatorname{rank}\big([T]_{B\to C}\big) = \dim \operatorname{Im} T$
- $[T_1]_{B\to C} = [T_2]_{B\to C} \Rightarrow T_1 = T_2$
- T is invertible $\iff$ $[T]_{B\to C}$ is invertible
- $S : W \to Z$, D basis of Z:
  $[S \circ T]_{B\to D} = [S]_{C\to D}\,[T]_{B\to C}$
- B' basis of V, C' basis of W:
  $[T]_{B'\to C'} = [I]_{C\to C'}\,[T]_{B\to C}\,[I]_{B'\to B}$
  - $\varphi : V \to V$, same basis both sides:
    $[\varphi]_{B'} = P^{-1}\,[\varphi]_B\,P$, where $P := [I]_{B'\to B}$

14. The Space $\mathcal{L}(V, W)$

$\mathcal{L}(V,W) := \{\varphi : V \to W \mid \varphi \text{ is linear}\}$
$\varphi, \psi \in \mathcal{L}(V,W)$, $\lambda \in \mathbb{F}$
B basis of V, $\dim V = n$; C basis of W, $\dim W = m$

- $(\varphi + \psi)(x) := \varphi(x) + \psi(x)$
- $(\lambda\varphi)(x) := \lambda\big(\varphi(x)\big)$
- $\varphi + \psi$ and $\lambda\varphi$ are linear
- $\mathcal{L}(V,W)$ is a vector space over $\mathbb{F}$
- $\varphi \mapsto [\varphi]_{B\to C}$ is an isomorphism $\mathcal{L}(V,W) \to \mathbb{F}^{m\times n}$
- $\dim V, \dim W < \infty$
  $\Rightarrow \dim \mathcal{L}(V,W) = \dim V \cdot \dim W = mn$

15. Dual Spaces, Functionals, and Annihilators

Linear functionals. A linear functional is a linear map $f : V \to \mathbb{F}$. It assigns a scalar to each vector and represents a linear measurement.

Dual space.

- $V^* := \mathcal{L}(V, \mathbb{F}) = \{f : V \to \mathbb{F} \mid f \text{ is linear}\}$
- V^* is a vector space
- $\dim V = n < \infty \Rightarrow \dim V^* = n$

Dual basis.
$B = \{v_1, \dots, v_n\}$ basis of V

- $B^* := \{f_1, \dots, f_n\}$, the unique functionals with $f_i(v_j) = \delta_{ij}$
- B^* is a basis of V^*, determined by B
- $\varphi = \sum_i \varphi(v_i)\,f_i$ for all $\varphi \in V^*$
- $f_i(v) = f_i\big(\sum_j \alpha_j v_j\big) = \alpha_i$, the i-th coordinate of $[v]_B$

Annihilator (MEAPES).
$S \subseteq V$

- $S^\circ := \{f \in V^* : f(s) = 0 \text{ for all } s \in S\}$
- S° is a subspace of V^*
- $S_1 \subseteq S_2 \Rightarrow S_1^\circ \supseteq S_2^\circ$
- $S^\circ = (\operatorname{span} S)^\circ$
- $\{S_i\}_{i\in I}$ subsets of V: $\big(\bigcup_i S_i\big)^\circ = \bigcap_i S_i^\circ$
- $W \leq V$, $\dim V < \infty$: $\dim W + \dim W^\circ = \dim V$

Pre-annihilator (TAT MERHAV HA EFASIM).
$G \subseteq V^*$

- $G_\circ := \{v \in V : f(v) = 0 \text{ for all } f \in G\} = \bigcap_{f\in G} \ker f$
- $G_\circ$ is a subspace of V
- $G_1 \subseteq G_2 \Rightarrow (G_1)_\circ \supseteq (G_2)_\circ$
- $G_\circ = (\operatorname{span} G)_\circ$
- $\{G_i\}_{i\in I}$ subsets of V^*: $\big(\bigcup_i G_i\big)_\circ = \bigcap_i (G_i)_\circ$
- $N \leq V^*$, $\dim V < \infty$: $\dim N + \dim N_\circ = \dim V$
- $\emptyset_\circ = V$

Difference between MERHAV HA EFES and MEAPES.

- MERHAV HA EFES: vectors in V mapped to zero by a linear map
- MEAPES: functionals in V^* that vanish on a subset

Second dual space.

- $V^{**} := (V^*)^*$
- canonical map $\Phi : V \to V^{**}$, $\Phi(v)(f) := f(v)$
  - *canonical*: defined with no choice of basis
- Φ is linear
- $\dim V < \infty \Rightarrow \Phi$ is an isomorphism
- $G \subseteq V^*$, $\dim V < \infty$: $\Phi(G_\circ) = G^\circ$ (in V^{**})
  - $\dim G_\circ = \dim G^\circ$

Dual (transpose) map.
$T : V \to U$

- $T^* : U^* \to V^*$, $T^*(f) := f \circ T$
- $\ker T^* = (\operatorname{Im} T)^\circ$
- $\operatorname{Im} T^* = (\ker T)^\circ$
- matrix form: $[T^*] = [T]^T$

16. Vandermonde Matrix

Given scalars $x_1, \dots, x_n \in \mathbb{F}$, the **Vandermonde matrix** is

$$V(x_1, \dots, x_n) := \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix}$$

Determinant.

$$\det V(x_1, \dots, x_n) = \prod_{1\leq i<j\leq n} (x_j - x_i)$$

Invertibility.

$$V(x_1, \dots, x_n) \text{ is invertible} \iff x_1, \dots, x_n \text{ are pairwise distinct}$$

Consequences.

- If $x_i = x_j$ for some $i \neq j$, then two rows are equal $\Rightarrow \det V = 0$
- If all x_i are distinct, then $\operatorname{rank}(V) = n$
- Appears naturally in polynomial interpolation

17. Permutations

A **permutation** of $\{1, \dots, n\}$ is a bijection

$\sigma : \{1, \dots, n\} \to \{1, \dots, n\}$.

The set of all permutations is denoted S_n.

Inversions. An inversion of σ is a pair (i, j) such that

$i < j \quad \text{and} \quad \sigma(i) > \sigma(j)$.

Let $\operatorname{inv}(\sigma)$ be the number of inversions of σ, sometimes denoted $S(\sigma)$.

Sign of a permutation.

$\operatorname{sgn}(\sigma) := (-1)^{\operatorname{inv}(\sigma)}$.

- σ is *even* if $\operatorname{sgn}(\sigma) = 1$
- σ is *odd* if $\operatorname{sgn}(\sigma) = -1$

Properties.
- $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau)$
- A transposition changes the sign
- $|S_n| = n!$

Determinant (Leibniz formula). For $A = (a_{ij}) \in \mathbb{F}^{n\times n}$,

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

Consequences.
- If two rows (or columns) are equal, every term cancels $\Rightarrow \det(A) = 0$
- Swapping two rows multiplies $\det(A)$ by -1
- Linearity of determinant follows from the formula

18. Block Matrices

Definition. A *block matrix* is a matrix partitioned into submatrices:

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix},$$

where the blocks have compatible sizes.

Block operations. If dimensions agree, block matrices behave like ordinary matrices:
- Addition and scalar multiplication are blockwise
- Matrix multiplication follows block multiplication rules

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} E & F \\ G & H \end{pmatrix} = \begin{pmatrix} AE + BG & AF + BH \\ CE + DG & CF + DH \end{pmatrix}$$

Block diagonal matrices.
If

$$A = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix},$$

$$\det(A) = \det(A_1)\det(A_2),$$
$$A \text{ invertible} \iff A_1, A_2 \text{ invertible}.$$

Block triangular matrices. If

$$A = \begin{pmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{pmatrix},$$

then

$$\det(A) = \det(A_{11})\det(A_{22}),$$
$$\operatorname{rank}(A) = \operatorname{rank}(A_{11}) + \operatorname{rank}(A_{22}).$$

Schur complement. Let

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \quad A_{11} \text{ invertible.}$$

The *Schur complement* of A_{11} in A is

$S := A_{22} - A_{21}A_{11}^{-1}A_{12}$.

- A invertible $\iff A_{11}$ and S invertible
- $\det(A) = \det(A_{11})\det(S)$

Block inverse formula. If A_{11} and S are invertible, then

$$A^{-1} = \begin{pmatrix} A_{11}^{-1} + A_{11}^{-1}A_{12}S^{-1}A_{21}A_{11}^{-1} & -A_{11}^{-1}A_{12}S^{-1} \\ -S^{-1}A_{21}A_{11}^{-1} & S^{-1} \end{pmatrix}.$$

Solving block systems. For

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} b \\ c \end{pmatrix},$$

solve:
1. $A_{11}x = b - A_{12}y$
2. $(A_{22} - A_{21}A_{11}^{-1}A_{12})y = c - A_{21}A_{11}^{-1}b$

Implementation tips.
- Use block structure to reduce computation
- Common in Gaussian elimination, LU decomposition, and numerical methods
- Essential for large sparse systems and matrix factorizations